MARKUS CECIL BASAÑES TECSON

EDUCATION

Master of Science in Chemistry (Concentration in Biochemistry) Summer 2026
San Francisco State University
Bachelor of Science in Biochemistry Spring 2025
San Francisco State University

RESEARCH EXPERIENCE

Staff Research Associate Fall 2026 - Present
CxT Discovery, Berkeley, CA

Graduate Student Researcher

San Francisco State University, San Francisco, CA
Principal Investigator: Eric R. Greene, Ph.D.

Oxidation Sensitive, Oligomeric State-Dependent Allosteric Switch in Human Glutamine Synthetase Spring 2025 - Fall 2026

- Led biochemical and structural investigation of human glutamine synthetase through the integration of single particle cryo-EM, biophysical assays, recombinant expression, purification through affinity and size-exclusion chromatography, enzyme kinetics, and molecular dynamics ensemble refinement.
- Identified a concentration-dependent activity change associated with destabilization of assembly status through the investigation of engineered variants at the conserved oligomeric interface.
- Collaborated with the Carroll Lab at San Jose State University to understand thermostability of engineered mutants which illustrated altered protein stability with the transition state analog bound compared to WT.
- Determined the structural basis of activity changes utilizing high resolution cryo-EM supplemented with single structure and MD ensemble refinement approaches highlighting the conformational dynamics of oligomeric-dependent regulation.

Mechanistic Insights into Human Glutamine Synthetase Disease Variants Spring 2025 - Spring 2026

- Supported the investigation of disease-associated glutamine synthetase mutations in humans using recombinant expression in E. Coli BL21 (DE3), protein purification, enzyme kinetics, and single structure refinement from resulting coulombic potential maps.
- Researched potential hGS-protein interactions with intrinsically disordered proteins, namely URI1, through recombinant expression, denaturing purification, and AC50 measurements.
- Provided mentorship to a fourth-year undergraduate in recombinant expression, affinity and size-exclusion chromatography for protein purification, biochemical assays, and data analysis.

Systematically Investigating Disease-Associated Mutations of Human Glutamine Synthetase Summer 2025

- Supported lab investigations in discovering how disease-associated mutations to human glutamine synthetase alter catalytic function through primer design, DNA extraction and purification, and expression in *E. coli* BL21(DE3).

Undergraduate Student Researcher

San Francisco State University, San Francisco, CA
Principal Investigator: Eric R. Greene, Ph.D.

Determining Allosteric Pathways in Human Glutamine Synthetase Fall 2023 - Spring 2025

- Collaborated with an undergraduate team to investigate the allosteric regulation of hGS using solved cryo-EM structures, cryo-EM data processing, molecular dynamic ensemble fitting, and graph theory.
- Utilized a Python-based workflow to process and connect cryo-EM ensemble data with graph theory methods through the deployment of CorrelationPlus which revealed a smaller network in the H145R, disease-associated mutant.
- Provided mentorship to several undergraduate students and led project meetings to help develop a scientific community, project understanding, and work towards project goals.

Measuring Residue Acid Dissociation and Molecular Interactions in Human Glutamine Synthetase Fall 2023 - Spring 2024

- Collaborated with an undergraduate team to analyze residue pKa values and electrostatic interactions in hGS using a combination of molecular visualization through ChimeraX and Python-based data analysis utilizing single structure reconstructions of WT decamer and filament.
- Identified conformational changes associated with ADP release and changes in the electrostatic environments.
- Revealed electrostatic features that may stabilize subunit interactions and contribute to allosteric regulation.

PUBLICATIONS

* denotes first author
‡ denotes mentored undergraduate

Research Articles and Manuscripts

1. Greene, E.R., Muniz, R., Yamamura, H., Hoff, S.E., Bajaj, P., **Tecson, M.C.B.**, Geluz, C.[‡], Lee, D.J., Thompson, E.M., Arada, A., Lee, G.M., Bonomi, M., Kollman, J.M., Fraser, J.S. (2026 October), *Product-stabilized filamentation by human glutamine synthetase allosterically tunes metabolic activity, eLife* **14**:RP108336, doi: https://doi.org/10.7554/eLife.108336.3
2. Yamamura, H.*, **Tecson, M.C.B.***, Martinez, K., Geluz, C.[‡], Hoff, S., Bazan, N., Leyva, R., Paprocki, M.J., Rascon, C., Schlauraff, J., Martel, A., Alkabanni, H., Bhattacharya, A., Fournier, D.R.[‡], Jr., Hales, M.[‡], Leung, R.[‡], Hernandez, E., Cruz, Y., Bonomi, M., Carroll, E.C., Greene, E.R. (Manuscript in preparation), *Oligomeric remodeling links active-site dynamics to activity in human glutamine synthetase.*
3. Mejia, E.*, **Tecson, M.C.B.***, Yamamura, H., Martinez, K., Gurung, L., Leung, R., Trinh, A., Bhattacharya, A., Jacobs R., Arada, A., Bautista, A.A., Bakhit, R.J., Bovell, T.N., Gutierrez, N.Y., Money, S.A., Nguyen, J., Robles, V., Rosello-Duenas, K.C., To, A.K.H., Vasquez-Portillo, E.S., Zaghoul, H.W., Thompson, E., Carroll, E.C., Greene, E.R. (Manuscript in preparation), *Mechanistic Insight into Disease Mutations of Human Glutamine Synthetase*.
4. Bautista, A., Geluz, C., Tozlovan, G.[‡], **Tecson, M.C.B.**, Greene, E.R. (Manuscript in preparation), *Structural basis for glutamine synthetase inhibition via adenylation.*

Perspectives and Reviews

1. **Tecson, M.C.B.*,** Geluz, C.*[‡], Cruz, Y., Greene, E.R. (2025 January), *Glutamine Synthetase: Diverse Regulation and Functions of An Ancient Enzyme*, *Biochemistry 2025, 64, 3, 547-554*. https://doi.org/10.1021/acs.biochem.4c00763

TEACHING AND MENTORING EXPERIENCE

Substitute Teacher Fall 2023 - October 2026

Napa Valley Unified School District, Napa, CA
Number of Students: ~180

- Facilitated a positive, inclusive, and affirming environment to help support a consistent learning experience for students.
- Adapted teaching methods to fit the needs of individual students.
- Effectively managed classrooms to encourage a positive learning environment.

Teaching Assistant Spring 2024

San Francisco State University, San Francisco, CA
Freshman Research Colloquium: 17 students

- Assisted in leading class discussions
- Mentored cohort in small groups, building on course objectives and class discussions
- Developed and led cohort-building activities to support and create a supportive learning environment and community

HONORS AND AWARDS

National Institutes of Health Master of Science (MS) Bridges to the Fall 2025 - Spring 2026

Doctorate Fellowship

Competitive NIH-funded fellowship providing **up to $36,000/year** in tuition, research support, and stipend.

Genentech Foundation Master of Science (MS) Scholar Spring 2025 - Fall 2025

Competitive two-year scholarship providing **approximately $34,000/year** in tuition, stipend, travel, and research support.

Michael & Greta McKinney Chemistry and Biochemistry Scholarship Spring & Fall 2024

Supports Undergraduate Students pursuing a degree in Chemistry or Biochemistry, providing $6,000 per semester.

National Institutes of Health Undergraduate Research Training Initiative for Student Enhancement (NIH U-RISE) Fellowship Spring 2024 - Spring 2025

Competitive undergraduate research fellowship providing approximately $28,000/year in tuition, research support, travel, and stipend.

CSUBIOTECH Student Travel Grant Spring 2024

Competitive $1,500 award to support travel to educational conferences

California State University Louis Stokes Alliance for Minority Participation in STEM (CSU-LSAMP) Fellowship Fall 2023 - Spring 2025

SELECTED PRESENTATIONS

* denotes presenter(s)
‡ denotes mentored undergraduate

Oral Presentations

Tecson, M.C.B.*, Geluz, C.‡, Fournier, D.R., Jr.‡, Hales, M.‡, Yamamura, H., Greene, E.R. (2025 October), *Functional consequences to oligomeric state changes in human glutamine synthetase*, 2025 American Chemical Society Western Regional Meeting, San Jose, CA, USA

Tecson, M.C.B.*, Geluz, C.‡, Fournier, D.R., Jr.‡, Hales, M.‡, Yamamura, H., Greene, E.R. (2025 March), *Oligomeric state-dependent activity changes in Human Glutamine Synthetase*, 2025 West Coast Structural Biology Workshop, Monterey, CA, USA

Tecson, M.C.*, Fournier, D.R., Jr.‡, Geluz, C.‡, Leung, R.‡, Hales, M.‡, Parmigiani, S., Yamamura, H., Greene, E.R. (2024 July), *Utilizing Ensemble-Function Relationships to Determine Allosteric Networks in Glutamine Synthetase*, San Francisco State University's 31st Annual Summer Undergraduate Research Symposium, San Francisco, CA, USA

Poster Presentations

Tecson, M.C.B.*, Yamamura, H., Geluz, C.J., Martinez, K., Hoff, S.E., Fournier, D.R., Hales, M., Leung, R., Hernandez, E., Cruz, Y., Bonomi, M., Carroll, E.C., Greene, E.R. (2026 February), *Functional consequences to oligomeric state changes in human glutamine synthetase*, 2026 Meeting of the Biophysical Society, San Francisco, CA, USA. Published in the Biophysical Journal 2026, 125 (4), 159a. 10.1016/j.bpj.2025.11.1101.

Tecson, M.C.B.*, Yamamura, H., Geluz, C.‡, Hales, M‡., Fournier, D.R., Jr.‡, Leung, R.‡, Greene, E.R. (2025 June), *Oligomeric-State Dependent Activity Changes in Human Glutamine Synthetase*, 39th Annual Meeting of the Protein Society, San Francisco, CA, USA

Tecson, M.C.*, Hernandez, E.*, Fournier, D.R., Jr.*, Cruz, Y.*, Yamamura, H., Greene, E.R. (2024 April), *Uncovering Ensemble Function Relationships of the Ancient Enzyme Glutamine Synthetase*, American Chemical Society Northern California Undergraduate Research Symposium, San Francisco, CA, USA

COMMUNITY INVOLVEMENT

Volunteer High School Chemistry Tutor & Mentor Fall 2022 - Fall 2026

Chemistry and Biochemistry Student Association Fall 2022 - Spring 2024

Public Relations Officer & Educational Outreach Chair

Associated Students of Napa Valley College Fall 2020 - Winter 2024

Senator-at-Large; President; Historical Advisor

Phi Theta Kappa Honor Society, Beta Beta Sigma Chapter Fall 2020 - Spring 2022

Vice President of Community Service Fall 2020 - Spring 2021